USER MANUAL

RPi500+ Task Manager

A Windows-style Task Manager for the Raspberry Pi 500+ (16 GB), with a Wi-Fi hotspot, visitor log, Bluetooth rescue and a local AI.



The Overview tab, running on a Raspberry Pi 500+

Watch everything	CPU, clock, temperature, memory, SSD, real input voltage and power draw at a glance
Take it anywhere	Creates its own Wi-Fi network, **RPI500-OPEN**, when there's no other network, and flashes the power light while it's on
See who's there	Live list of everyone on your hotspot, with a visitor log saved to the Desktop
Never get stuck	Bluetooth rescue from the keyboard, VNC remote access, and a local AI that works offline

Version 1.0 · September 2026 · For Raspberry Pi OS Bookworm or newer

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CHAPTER 1

Welcome

RPi500+ Task Manager puts everything about your Pi on one screen, the way Task Manager does on Windows. It adds tools for taking the Pi on the road: its own Wi-Fi network, a visitor log, a power-light signal, remote access that always works, and a small AI that needs no internet.

What it can do

- **Overview:** gauges for CPU, clock speed, temperature, memory, swap and disk, plus a 60-second CPU graph, per-core bars, core voltage and power-warning lights.
- **Processes:** a sortable list of everything running, with End task and Force kill.
- **Network:** blinking LINK, RX and TX lights for Ethernet and Wi-Fi, speeds, totals and a traffic graph.
- **Tools:** one-press memory cleanup, Bluetooth rescue and pairing, the hotspot with a join-by-QR-code, battery-saver CPU mode, overclock presets, VNC, and reboot or shut down.
- **Who's Connected:** a window listing every device on your hotspot, with a visitor log that can save itself to your Desktop.
- **AI:** an on/off switch and a chat with a small AI model that runs entirely on the Pi.
- **Always reachable:** if the Pi powers up with no network, it creates **RPI500-OPEN** by itself, and VNC is always on, so you can remote in with no screen. Switch on **Tailscale** and you can reach it from anywhere with internet, too.

What you need

Item	Details
A Raspberry Pi	Built for the Pi 500+ with 16 GB. It adapts to the Pi 500, Pi 5, Pi 400 and Pi 4 too, showing only what each one can measure.
Operating system	Raspberry Pi OS **Bookworm or newer** with the desktop. The AI tab needs the **64-bit** version.
Internet (once)	Needed during installation to fetch a few packages and the AI model. After that, everything works offline.
Power	The official 27 W (5 V 5 A) USB-C supply, or a good USB-C battery bank (see chapter 12).

CHAPTER 2

Installing

Installation is one command. It takes a few minutes, most of it downloading the AI model.

Step by step

1. Copy **rpi500-taskmgr.zip** to the Pi (a USB stick works) and double-click it to unzip it, or right-click it and choose *Extract Here*.
2. Open the **rpi500-taskmgr** folder, then press **F4** (or choose *Tools → Open Current Folder in Terminal*) to open a terminal there.
3. Type the following and press Enter. Give your password if it asks:

```
sudo bash install.sh
```

1. When it prints **Done!**, restart the Pi:

```
sudo reboot
```

After the restart, the Task Manager opens by itself. You can also open it anytime with `Ctrl+Shift+Esc`, or from *Menu → System Tools → RPi500+ Task Manager*.

What the installer sets up

Setup	What it means for you
The app	Adds a menu entry and opens the Task Manager when the desktop starts.
No password prompts	The buttons that need admin rights (clean memory, hotspot and so on) just work.
RPI500-OPEN hotspot	Password **fifty-seven57**. The Pi is at **192.168.1.250**.
Auto-hotspot at boot	If the Pi hasn't joined any network about 60 seconds after powering on, it starts the hotspot by itself.
Flashing power light	The power light blinks the whole time the hotspot is on.
VNC on port 5900	Remote desktop access. The Pi also starts straight into the desktop, so VNC works with no monitor attached.
Keyboard lighting	Adds Raspberry Pi's **rpi-keyboard-config** tool so the keys can glow while the hotspot is on.
Global shortcuts	`Ctrl+Shift+Esc` opens the Task Manager and `Ctrl+Alt+B` opens Bluetooth pairing, from anywhere. This works on every Raspberry Pi OS desktop (labwc, Wayfire and X11).

Setup	What it means for you
Web page (optional)	Ready to switch on from the Tools tab, so phones and PCs on your network can use the Task Manager in a web browser.
Tailscale (optional)	Ready to switch on from the Tools tab whenever you want to reach the Pi from anywhere. With an auth key (see below) it joins straight away.
Local AI	Installs the Ollama engine and the **llama3.2:3b** model, if the Pi is online and running 64-bit Raspberry Pi OS.

Install options

Type any of these right after sudo to change what the installer does. For example:

```
sudo SKIP_AI=1 WIFI_COUNTRY=GB bash install.sh
```

Option	Default	What it does
`WIFI_COUNTRY=US`	US	The Wi-Fi country, which the hotspot needs. Only used if none is set yet.
`SKIP_AUTOLOGIN=1`	off	Don't make the Pi start straight into the desktop.
`VNC_RESOLUTION=1920x1080`	1920x1080	Screen size VNC uses when no monitor is plugged in.
`TAILSCALE_AUTHKEY=tskey-…`	off	Join your Tailscale network during the install (chapter 11).
`WEB_PASSWORD=…`	off	Turn on the web page during the install, with this control password (chapter 11).
`WEB_PORT=8090`	8090	The port the web page uses.
`SKIP_AI=1`	off	Don't install the local AI.
`AI_MODEL=gemma3:4b`	llama3.2:3b	Which AI model to download.

Updating and removing

To update to a newer version, unzip it and run the installer again. Your settings, visitor log and AI models are kept.

To remove the Task Manager, run the command below in the same folder. It leaves VNC, desktop auto-login and any overclock alone, so set the overclock back to Stock first if you changed it. The AI engine and its models also stay, and the uninstaller prints the command to remove those.

```
sudo bash uninstall.sh
```

CHAPTER 3

The main window

Five tabs sit under a top bar that's always visible, and a status bar runs along the bottom.

The top bar

Item	What it shows
Username	The Pi user account the Task Manager is running as, for example **pi**.
IP address	The Pi's current network address or addresses, such as **192.168.1.250** when it's running the hotspot.
People counter	How many devices are on your hotspot, for example **4 on hotspot**. It turns green when anyone is connected. Click it to open the Who's Connected window.
eth0 / wlan0 lights	**LINK** is green when connected and red when not. **RX** blinks green while data comes in, and **TX** blinks amber while data goes out.
Bluetooth	Green with the device name when something is connected. Amber means Bluetooth is on but nothing is connected, and red means Bluetooth is off.

Tabs

Tab	Key	Use it for
Overview	F1	The health of the Pi: load, heat, memory, disk and power
Processes	F2	Seeing what's running and closing programs
Network	F3	Connections, speeds and hotspot devices
Tools	F4	Buttons for memory, Bluetooth, hotspot, performance and remote access
AI	F5	Chatting with the local AI

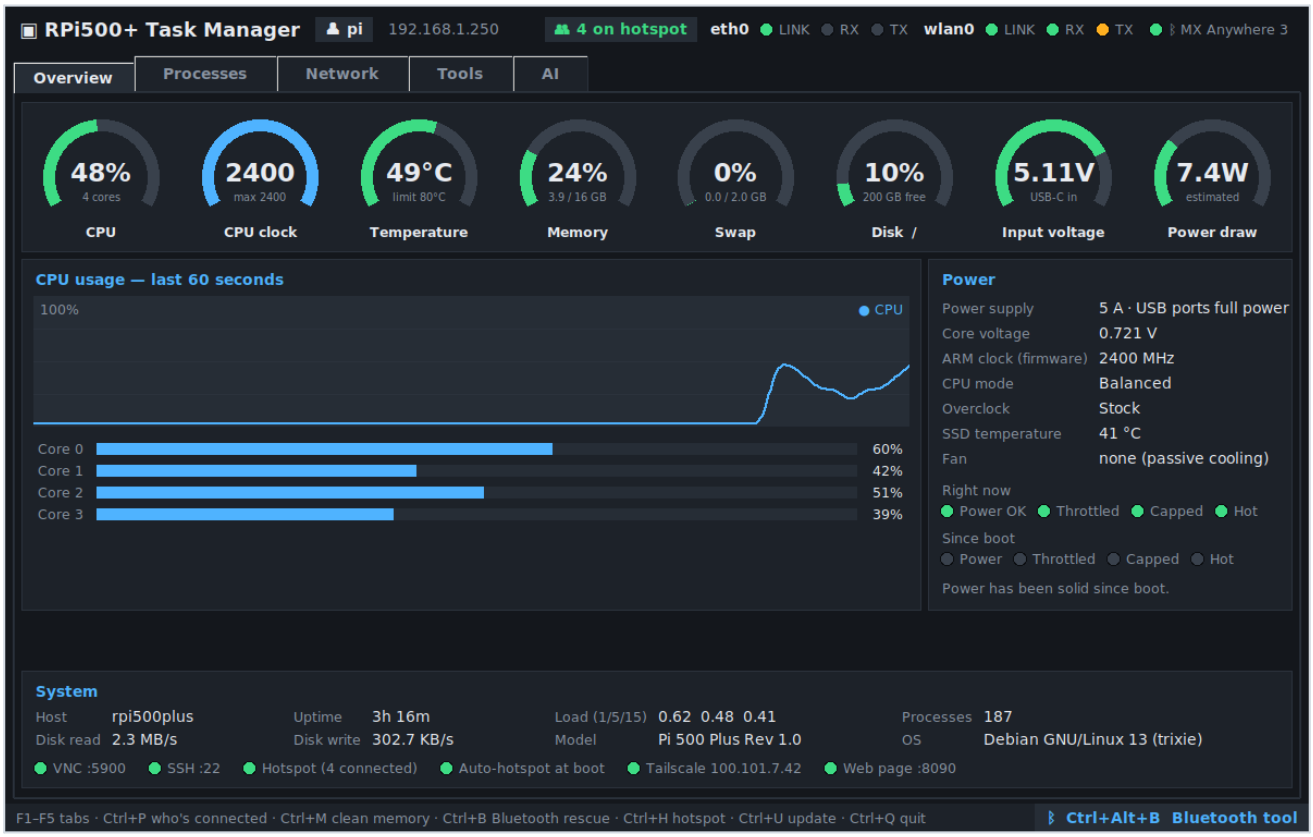
`Alt+1` to `Alt+5` also switch tabs. The status bar at the bottom shows the main shortcuts, and then the result of whatever you last did. Its right end always shows **Ctrl+Alt+B Bluetooth tool** as a reminder for when your mouse stops working. Clicking it opens the Bluetooth window too.

> **Tip.** Every part of the app works from the Pi 500+'s keyboard alone. Use Tab and the arrow keys to move, and Space or Enter to press a button or flip a switch. See chapter 13 for every shortcut.

CHAPTER 4

Overview tab

Everything about the Pi's health on one page.



Overview tab

Gauges

Each gauge is green when things are fine, amber when they're getting high, and red when you should pay attention.

Gauge	What it measures	Amber / red at
CPU	How busy the processor is, across all 4 cores	70% / 90%
CPU clock	Current speed in MHz (the maximum is 2400 on a stock Pi 500+)	blue, no warning
Temperature	Processor temperature. The Pi slows itself down at 80 °C	65 °C / 80 °C
Memory	RAM in use, out of the total	75% / 90%
Swap	Overflow memory on the SSD. With 16 GB it's rarely used	50% / 80%

Gauge	What it measures	Amber / red at
Disk /	How full the built-in 256 GB SSD is, with free space shown	80% / 92%
Input voltage	The real voltage arriving at the USB-C socket, measured by the Pi's power chip. Healthy is about 5.0–5.2 V	below 4.95 V / 4.8 V
Power draw	An estimate of the whole Pi's power use in watts. About 3 W idle, and 8–12 W flat out	15 W / 21 W

CPU graph and cores

The graph shows total CPU use over the last 60 seconds. The bars underneath show each of the 4 cores, so you can spot a single program maxing out one core.

Power panel

Power supply shows how much current your charger or battery bank offered when the Pi started. At **5 A** the USB ports get full power. At **3 A**, which most battery banks give, the Pi runs fine but the USB ports share only 600 mA. That's plenty for a keyboard, mouse or USB stick, but not for a USB hard drive. **SSD temperature** is the built-in drive's own reading, and it's comfortable below about 60 °C. **Fan** says *passive cooling* because the Pi 500+ uses a large internal heatsink instead of a fan.

Core voltage is what the processor is running at, not the battery voltage. **ARM clock** is the speed the firmware reports, **CPU mode** is the performance setting from the Tools tab, and **Overclock** shows whether a speed boost is set.

Under those are two rows of lights. **Right now** is green when all is well and red when a problem is happening at this moment. **Since boot** turns amber if the problem happened at any point since the Pi started.

Light	Meaning
Power	The supply voltage dropped too low. Almost always a weak or nearly empty battery bank, or a long or thin cable.
Throttled	The Pi slowed itself down to protect itself.
Capped	The top speed was limited.
Hot	The processor reached its temperature limit.

Heads up. The **Input voltage** gauge is your battery-bank fuel warning. It sags as a bank empties or when a cable is too thin. When it turns amber, plan to recharge. The Pi can't read a battery bank's own charge level, only the voltage it delivers.

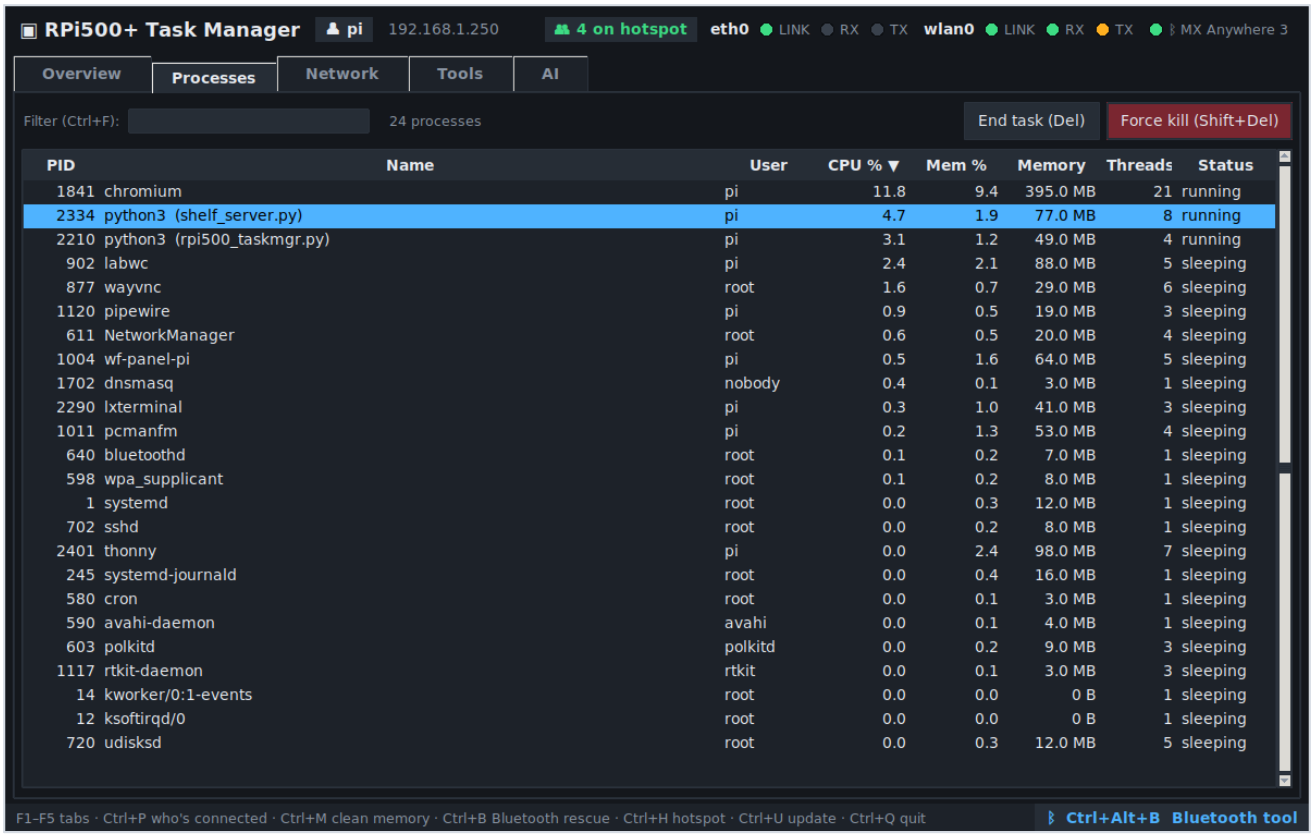
System panel

The name of the Pi, how long it's been on, the load average (1, 5 and 15 minutes), how many processes are running, disk read and write speed, the model and the operating system. The lights along the bottom show whether **VNC** and **SSH** are running, whether the **hotspot** is on (and how many are connected), and whether **auto-hotspot at boot** is enabled.

CHAPTER 5

Processes tab

Every program and background service that's running, like the Details tab on Windows.



Processes tab, sorted by CPU use. The selected row is the one End task will close.

Column	Meaning
PID	The process's ID number
Name	The program. For Python and other scripts, the script name is shown in brackets.
User	Who it's running as. **pi** (or your username) means your own programs, and **root** means system services.
CPU %	Share of the whole processor (all 4 cores together)
Mem % / Memory	Share of RAM, and the actual amount used
Threads, Status	How many threads it runs, and whether it's running or sleeping

Using it

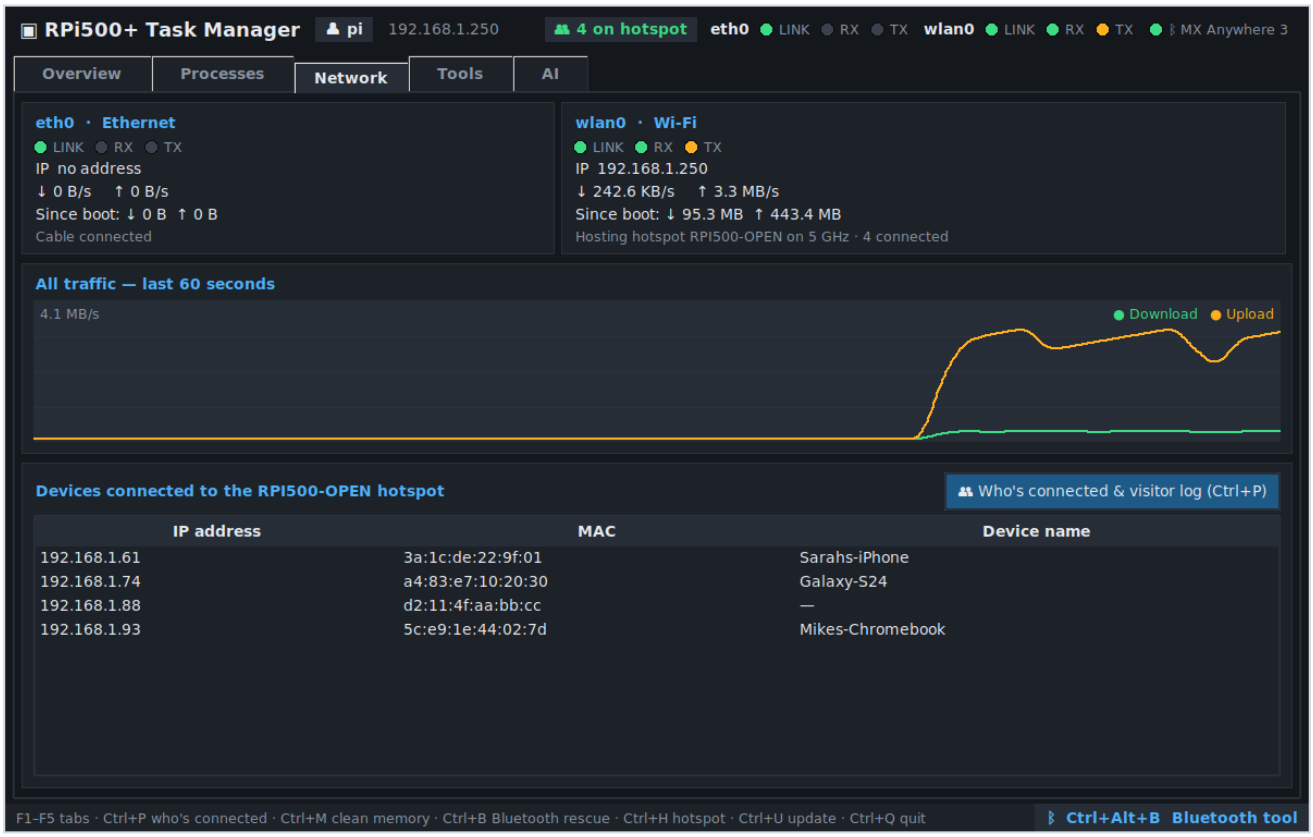
- **Sort** by clicking a column heading. Click it again to reverse the order. The arrow shows the current sort.
- **Find** a program with `Ctrl+F`, then type part of its name or PID. `Esc` jumps back to the list.
- **End task** (`Del`) politely asks the program to close. It's the one to try first.
- **Force kill** (`Shift+Del`) stops it immediately, for programs that are frozen.
- Both ask you to confirm. System processes are handled for you with admin rights.

> **Heads up.** Ending system processes such as **labwc** (the desktop), **NetworkManager** or **wayvnc** can close your desktop, cut the network or drop your VNC session. If that happens, a reboot fixes it.

CHAPTER 6

Network tab

Live traffic on every network connection.



Network tab while the Pi is hosting the hotspot for four devices

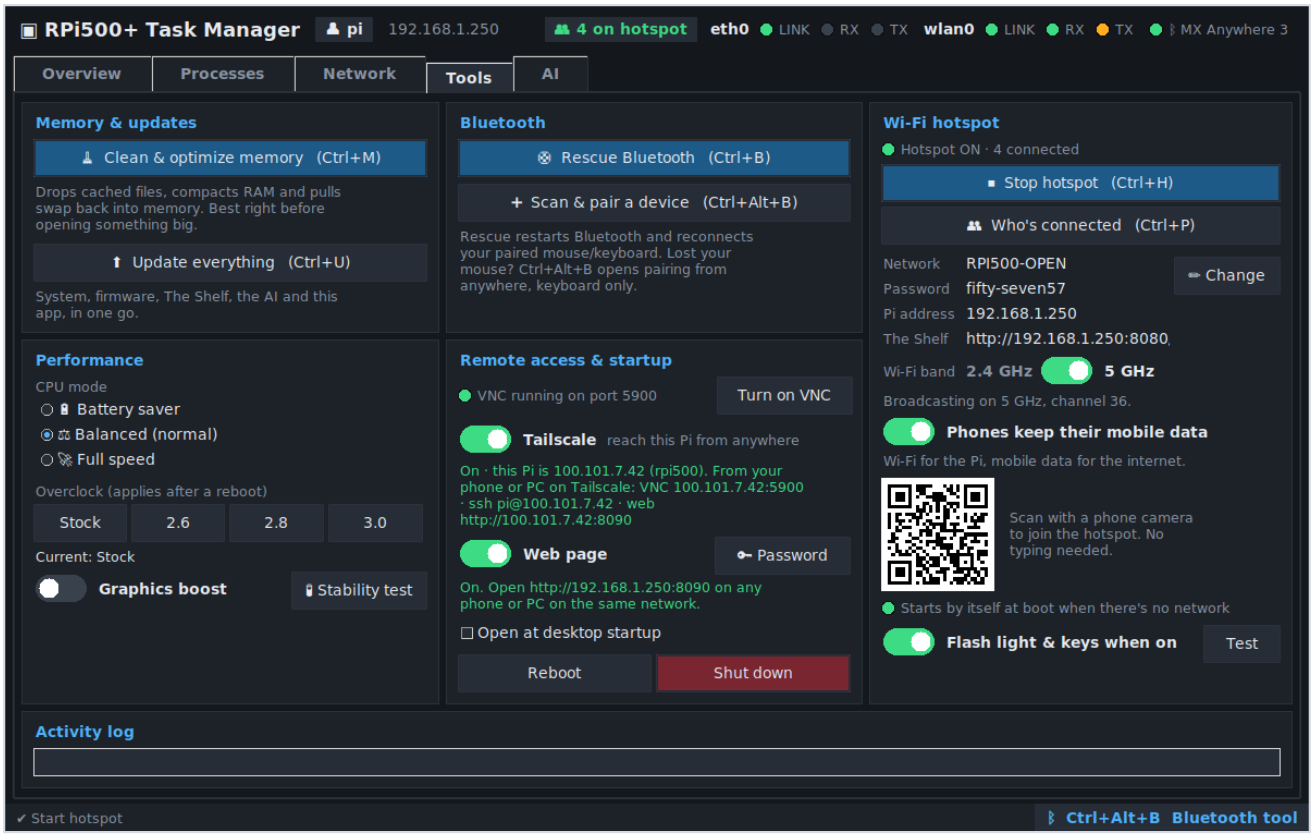
- **A card for each connection** (eth0 is Ethernet, wlan0 is Wi-Fi) with LINK, RX and TX lights, the IP address, current download and upload speed, and totals since boot.
- **The Wi-Fi card's bottom line** says whether the Pi has joined a network (with signal strength: excellent, good, fair or weak) or is hosting RPI500-OPEN, and how many devices are connected.
- **The traffic graph** shows download (green) and upload (amber) across all connections for the last minute.
- **The device list** shows everyone on the hotspot. The button opens the full Who's Connected window (chapter 8).

> **Tip.** When people download photos or programs from The Shelf, that's **upload** from the Pi's point of view, so the amber line will be the busy one.

CHAPTER 7

Tools tab

Buttons for everyday fixes and settings. Results appear in the Activity log at the bottom.



Tools tab

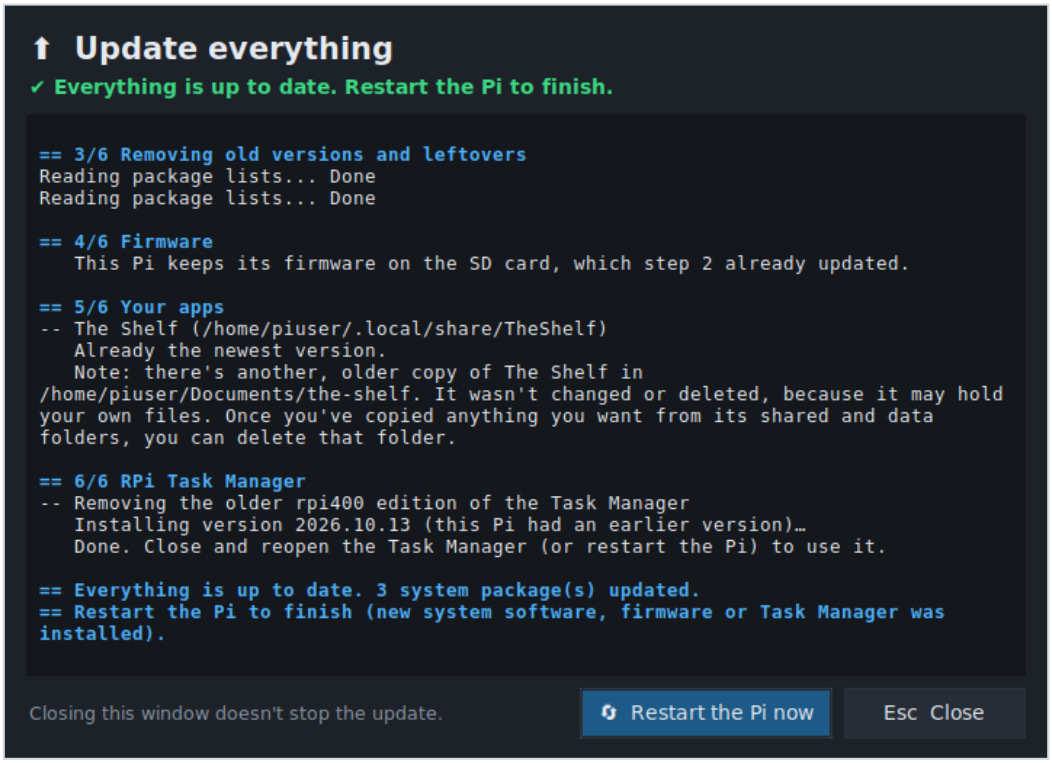
Memory

Clean & optimize memory (`Ctrl+M`) clears the file cache, tidies RAM and pulls data back from swap if there's room. The log shows how much memory was freed. Linux fills spare memory with cache on purpose and refills it over time, so it's most useful just before opening something big.

Update everything

⬆ Update everything (`Ctrl+U`) brings the whole Pi up to date in one go, so you never have to remember the commands. Confirm, and a window shows each step as it happens.

Step	What it updates
1–2 System software	Raspberry Pi OS and every installed program, including Tailscale
3 Old versions	Removes old kernels, packages nothing needs any more, and downloaded installer files
4 Firmware	The Pi's boot firmware (on boards that keep it on a chip)
5 Your apps	The local AI engine (Ollama), if installed, and **The Shelf**: its program files are replaced with the newest version from GitHub, and your pictures, music, notes and other files are kept. If The Shelf was running, it restarts with the new version.
6 RPi Task Manager	Installs a newer version of this app from GitHub if there is one, and removes any other edition of it (for example a Pi 400 copy on a Pi 500+) so there's only one.



The update window after a finished update, offering to restart the Pi

- It needs internet, so it won't run while the Pi is only on its own hotspot. It takes 5 to 30 minutes; keep the Pi plugged in. You can carry on using the Pi meanwhile.
- Closing the window doesn't stop the update. Press the button again to watch it, and the line under the button shows when the last update finished.
- When new system software, firmware or a new Task Manager arrived, it asks you to **restart the Pi**.
- If it finds a second copy of The Shelf somewhere else, it leaves that copy alone, since it may hold your own files, and tells you where it is so you can delete it.

Bluetooth

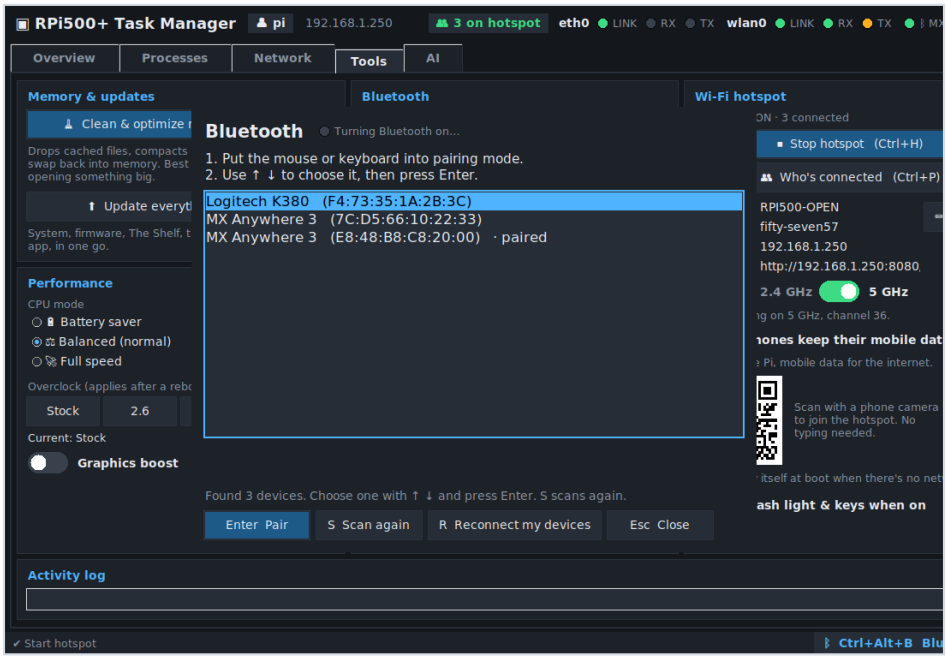
- **Rescue Bluetooth** (`Ctrl+B`) restarts Bluetooth and reconnects your paired mouse and keyboard. Use it when a device you've already paired stops working. It takes about 15 seconds. Press a key or click on the device to wake it up.
- **Scan & pair a device** (`Ctrl+Alt+B`) opens the Bluetooth window described below.

Lost your mouse? Ctrl+Alt+B

Press `Ctrl+Alt+B` on the Pi 500+'s keyboard from **anywhere**, even with the Task Manager closed. A Bluetooth window opens, already selected, so you never need a mouse. It turns Bluetooth on and scans for 10 seconds:

1. Put the new mouse or keyboard into pairing mode (usually by holding its connect button until a light flashes).
2. Use `↑` `↓` to choose it. New devices are listed first, and ones you've paired before are marked *paired* at the bottom.
3. Press `Enter` . If a large code appears, type it on the **new** keyboard and press its Enter key. Other devices pair by themselves.

Key	What it does
`Enter`	Pair the chosen device
`S`	Scan again
`R`	Reconnect my devices: restarts Bluetooth and reconnects everything you've paired before
`Esc`	Close the window



The Ctrl+Alt+B Bluetooth window, showing the code to type on a new keyboard

Wi-Fi hotspot

Start or stop RPI500-OPEN (`Ctrl+H`), open the Who's Connected window (`Ctrl+P`), and show the network name, password, the Pi's address and The Shelf's address. The **Wi-Fi band** switch chooses 2.4 GHz or 5 GHz, and the line under it shows the band and channel in use. ✏ **Change** sets your own network name and password, and **Phones keep their mobile data** lets phones on the hotspot stay online through their own data. The QR code lets people join by pointing their phone camera at it. Chapter 9 covers the hotspot in detail.

Performance

CPU mode	What it does
Battery saver	Keeps the CPU at its lowest speed. Cooler, and makes a battery last longest.
Balanced (normal)	Speeds up only when needed. The Pi's normal setting.
Full speed	Always at top speed. The most responsive, but uses the most power.

CPU mode resets to Balanced after a reboot. **Overclock** sets a permanent speed boost that takes effect after a reboot, and the app offers to reboot for you:

Button	Speed	Notes
Stock	2.4 GHz	The Pi 500+'s normal speed. Use this to undo an overclock.
2.6	2.6 GHz	A mild boost with a tiny voltage bump.
2.8	2.8 GHz	Works on most boards. Runs warmer and draws more power.
3.0	3.0 GHz	The edge of what these chips do, with a bigger voltage bump. Many boards manage it, some don't. Always run the Stability test after choosing it.

Heads up. If the Pi won't start after an overclock, take out its SSD (or SD card, if you boot from one) and connect it to another computer with an adapter. Open `config.txt` on the boot drive, and delete the lines between the two `rpi500-taskmgr overclock` markers at the end. A backup of the original is saved as `config.txt.taskmgr-backup`.

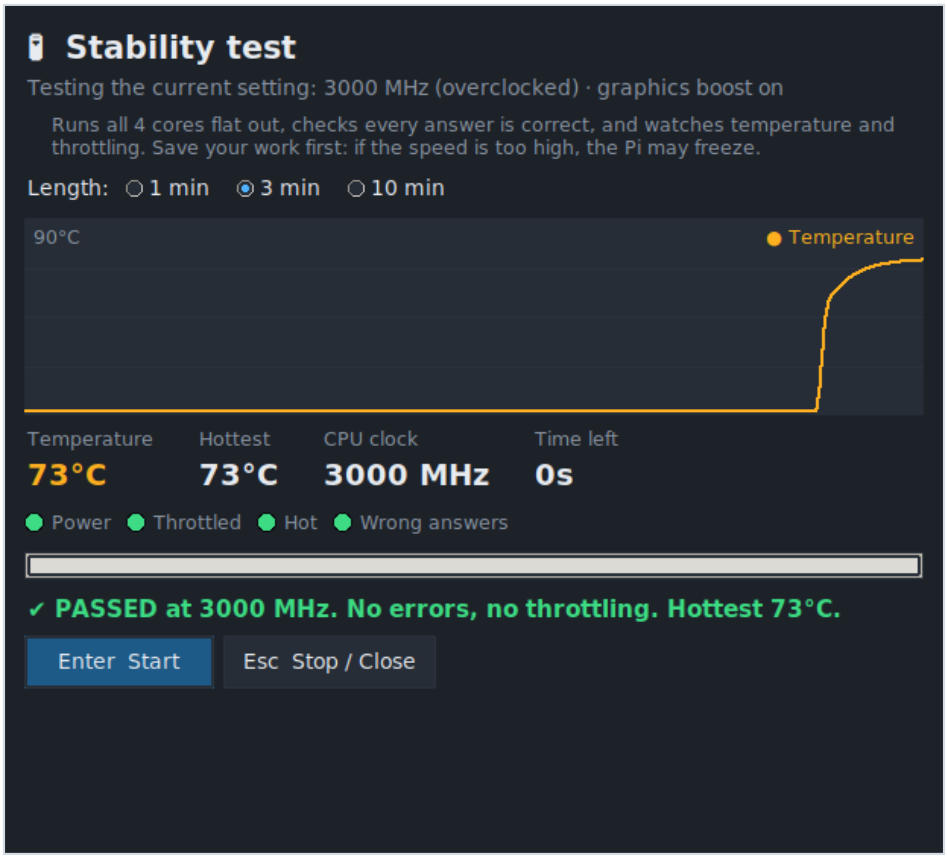
Graphics boost

The **Graphics boost** switch under the overclock buttons raises the graphics chip from its normal 910 MHz to 1000 MHz. That makes the desktop, video and browser graphics a little smoother. It's independent of the CPU speed, takes effect after a reboot, and the app offers to reboot for you.

Stability test

Press **Stability test** after choosing a faster speed and rebooting. Pick 1, 3 or 10 minutes, then press `Enter`. It runs all four cores flat out, checks that every calculation comes out right, and watches the temperature, clock speed and the Pi's own warning flags. A live graph shows the temperature as it climbs. `Esc` stops it early.

Result	What it means
✔ **PASSED**	No errors and no slowing down. This speed is good on your board.
⚠ **STABLE, BUT HOT**	No errors, but it got hot enough to slow itself down. Short bursts get the full speed, long heavy jobs won't. That's normal for the fan-less 500+ at 3.0 GHz.
⚠ **POWER PROBLEM**	The supply dipped under load. Use the 27 W supply or a stronger battery bank before trusting this speed.
✖ **FAILED**	The CPU got wrong answers. This speed isn't stable on your board. Choose a lower speed and reboot.



A 3-minute test at 3.0 GHz with graphics boost: passed, peaking at 73 °C

Heads up. Save your work before testing. If a speed is too high, the Pi may freeze or restart partway through. If that happens, the Task Manager notices the next time it starts and tells you which speed caused it, so you know to step down.

Remote access & startup

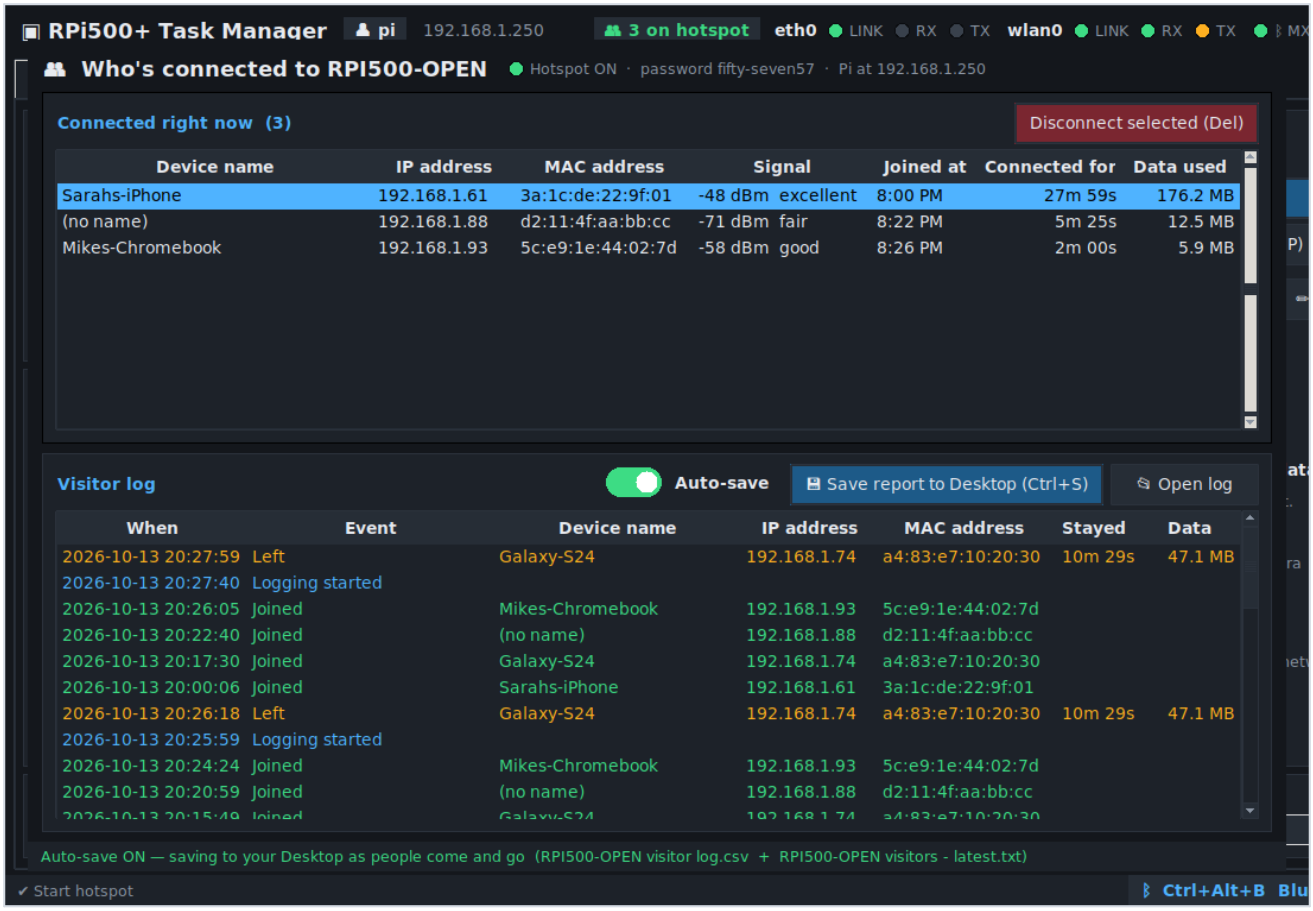
- The VNC light shows whether remote desktop is running on port 5900. **Turn on VNC** enables it if it's off.
- The **Tailscale** switch lets you reach this Pi from anywhere, even on mobile data. Under the switch you'll see the Pi's Tailscale address and exactly what to type for VNC and SSH. Chapter 11 walks through the one-time sign-in.
- The **Web page** switch lets any phone, tablet or PC on your network use the Task Manager in a web browser. **Password** sets or changes the password its controls need. Chapter 11 has the details.
- **Open at desktop startup** controls whether the Task Manager opens itself when you log in.
- **Reboot** and **Shut down** both ask for confirmation first.

CHAPTER 8

Who's Connected & the visitor log

See everyone on your RPI500-OPEN network, and keep a record of who came and went.

Open it by clicking the people counter in the top bar, by pressing `Ctrl+P`, or with the **Who's connected** button on the Network or Tools tab.



The Who's Connected window. Joins are green, leaves are amber, and hotspot events are blue.

Connected right now

Each device shows its name, IP address, MAC address (its hardware ID), signal strength, when it joined, how long it's been on, and how much data it has used. To disconnect someone, select them and press `Del`. They can rejoin if they still have the password.

Visitor log

Every join and leave, newest first. Leaves show how long the device stayed and how much data it used. **Logging started** marks when the Task Manager began watching.

Saving to the Desktop

Control	What it does
Auto-save switch (`Ctrl+A`)	**On** (the default): every join and leave goes straight into `RPI500-OPEN visitor log.csv` on your Desktop. That file opens like a spreadsheet in LibreOffice Calc and keeps growing across days and reboots. `RPI500-OPEN visitors - latest.txt` is also kept up to date, so the Desktop always shows the current picture. **Off:** nothing is written to the Desktop. The window still records everything, and the bar at the bottom counts events waiting to be saved. Turn it back on or press Save, and they're written. Nothing is lost. The switch position is remembered.
Save report to Desktop (`Ctrl+S`)	Saves a snapshot called `Hotspot visitors YYYY-MM-DD HHMM.txt` listing who's on now, the recent activity, and how many different devices visited.
Open log	Opens the spreadsheet log.

> **Tip.** Device names come from what each phone or laptop reports about itself, so some show as **(no name)**. Many phones also use a random hardware ID for privacy, so the same phone can appear as a new device on a different day. Logging happens while the Task Manager is running. It starts itself with the desktop, so that's normally all the time.

CHAPTER 9

The hotspot, anywhere

How RPI500-OPEN works, and how people connect to The Shelf.

Setting	Value
Network name	**RPI500-OPEN** unless you change it
Password	**fifty-seven57** unless you change it
Pi's address	**192.168.1.250**
The Shelf	**http://192.168.1.250:8080/**
Remote desktop	**192.168.1.250:5900** (VNC)

It starts by itself

Turn on the Pi somewhere with no known Wi-Fi and no Ethernet cable. About **60 seconds** later it creates RPI500-OPEN. You don't need a screen, a keyboard or the Task Manager for this. You can also start or stop it anytime with `Ctrl+H`. The Overview tab's **Auto-hotspot at boot** light shows this is enabled.

2.4 GHz or 5 GHz

The **Wi-Fi band** switch in the Tools tab's hotspot box picks which band RPI500-OPEN broadcasts on.

	2.4 GHz (default)	5 GHz
Range	Longest, and goes through walls better	Shorter, best in the same room
Speed	Slower, and crowded where there's lots of Wi-Fi	Noticeably faster
Devices	Every phone, laptop and gadget	Most modern devices. Some older or cheap ones can't see it
Channel	6	36

- If the hotspot is running, switching restarts it. Connected devices drop off for a few seconds and rejoin by themselves, but a VNC session over the hotspot needs reconnecting.
- Your choice is remembered, including for the automatic hotspot at boot.
- **You can't get locked out:** if 5 GHz won't start somewhere (local Wi-Fi rules, or the chip refuses), the Pi falls back to 2.4 GHz by itself, and the line under the switch tells you it did.
- Pick 2.4 GHz for a room full of mixed devices or a big space, and 5 GHz when a few people nearby want fast downloads from The Shelf.

How many people can join?

Roughly **8 to 10 devices at once** on the Pi 500+'s built-in Wi-Fi. The limit comes from the Wi-Fi chip's own software, so it can vary with system updates. To find yours, have people join one at a time while watching the people counter. Everyone also shares the Wi-Fi speed, so a handful of people browsing The Shelf is comfortable, while many large downloads at once will be slow.

The flashing power light and glowing keyboard

While the hotspot is on, the power light blinks about once a second **and every key on the keyboard breathes blue**, so you can tell from across the room that the Pi is broadcasting. Your own keyboard lighting comes back exactly as it was the moment the hotspot stops. The moment the hotspot stops, the light goes back to normal. This works even with the Task Manager closed.

- Turn it on or off with the **Flash light & keys when on** switch in the Tools tab. **Test** flashes it for 6 seconds.
- The power light's red and green parts both flash, so it may look amber.
- While it flashes, the light no longer shows SD-card activity. That comes back when the hotspot stops.

Letting people connect

1. On their phone or laptop, they join **RPI500-OPEN** with password **fifty-seven57**, or scan the QR code on the Tools tab.
2. They open a web browser and go to **http://192.168.1.250:8080/** to reach The Shelf.
3. You watch them arrive in the Who's Connected window.

Phones keep their mobile data

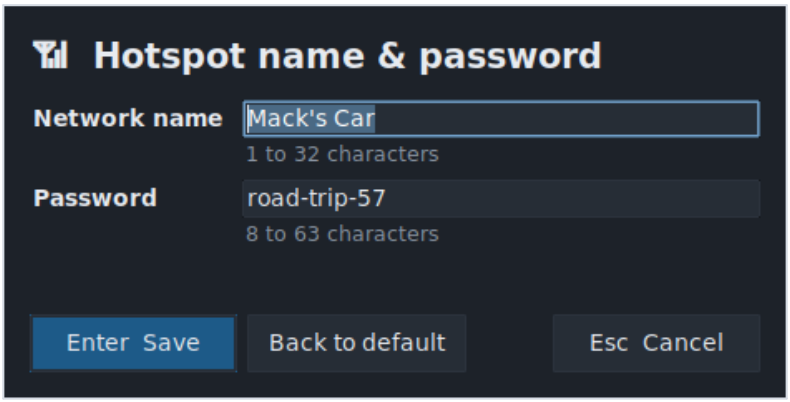
The hotspot has no internet of its own. Normally a phone joined to Wi-Fi sends *everything* over it, so apps like Teams, messages and maps go quiet while you listen to The Shelf. The **Phones keep their mobile data** switch in the Tools tab fixes that, and it's on from the start.

- **On:** the hotspot gives phones an address but no route to the internet. Each phone then uses Wi-Fi only to reach the Pi and The Shelf, and its own mobile data for everything else. You get both at once.
- **Off:** devices send all their traffic through the Pi. Use this only when the Pi itself is online, for example with an Ethernet cable, and you want to share that connection.
- Changing it restarts the hotspot. Devices rejoin by themselves after a few seconds.
- If a phone still acts offline, forget the network on it and join again, so it picks up the new setting. On Android, if it asks *This network has no internet access. Stay connected?*, choose **Yes** and tick *Don't ask again*. Turning on *Mobile data always active* in Android's Developer options also helps.

- Mobile data is used only for the internet. Music from The Shelf still comes from the Pi over Wi-Fi, so it doesn't use up your data plan.

Changing the name and password

Press ✏ **Change** next to the network name in the Tools tab. Type a new **Network name** (1 to 32 characters) and **Password** (8 to 63 characters), then press `Enter` to save, or `Esc` to cancel. **Back to default** fills in RPI500-OPEN and fifty-seven57 again.



Choosing your own hotspot name and password

- The QR code and everything the app shows update straight away, and the automatic hotspot at boot uses the new name too.
- If the hotspot is on, it restarts with the new name. Everyone has to join again with the new password, including you, if you're using VNC over the hotspot.
- Your phone may still remember the old name. That's harmless, and you can forget it in the phone's Wi-Fi settings.

Coming home

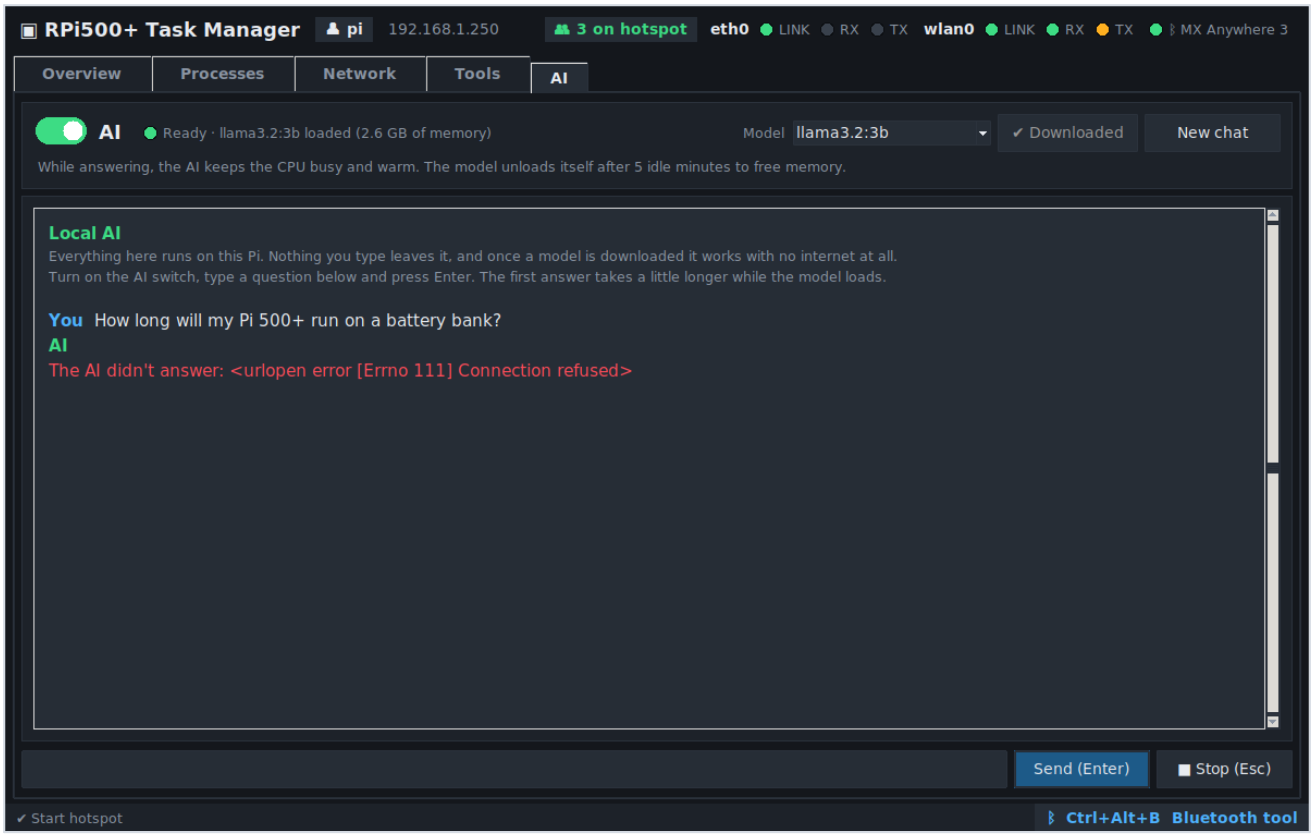
Once the hotspot is on, the Pi stays on it. When you're back in range of your home Wi-Fi, press `Ctrl+H` (**Stop hotspot**) and the Pi reconnects to your usual network.

> **Heads up.** Anyone with the password can join, and from there they can reach the VNC login screen. VNC still needs your Pi username and password, so make sure that password is a strong one. Share the hotspot password only with people you're happy to have on it.

CHAPTER 10

AI tab

Chat with a small AI that runs entirely on your Pi. Nothing you type leaves it, and it works with no internet.



The AI tab answering a question, with its speed shown underneath

Getting started

1. Flip the **AI** switch on. It takes a few seconds to start, and the light turns green when it's ready.
2. Type a question in the box at the bottom and press `Enter`.
3. The answer appears word by word. The first answer takes a little longer while the model loads into memory.

If the switch says the AI engine isn't installed, flip it anyway. The app offers to install it, which needs the internet and a few minutes, and then downloads the default model.

Controls

Control	What it does
AI switch	Starts and stops the engine. Off means it uses no memory or CPU, which is ideal on battery.
Model list	Chooses which AI model answers. Models not yet on the Pi say so underneath.
Download	Fetches the chosen model. It needs internet once, and then it's stored on the Pi.
New chat	Clears the conversation so the AI forgets the earlier questions.
Stop (`Esc`)	Stops an answer partway through.

Choosing a model

Model	Size	Best for
qwen2.5:1.5b	about 1 GB	The fastest answers, good for quick questions.
llama3.2:3b	about 2 GB	The default. A good balance of speed and smarts.
gemma3:4b	about 3.3 GB	Smarter, a little slower.
qwen2.5:7b	about 4.7 GB	The smartest, but slow: a few words a second.

Under each answer the app shows how fast it was written, in **tokens per second** (a token is roughly a short word). The model unloads itself after 5 idle minutes to free memory.

> **Heads up.** These tiny models are handy for quick questions, explanations and drafting, but they know far less than big online AIs and can be confidently wrong. Double-check anything important. While answering, the AI works the CPU hard, so the Pi warms up and a battery drains faster.

CHAPTER 11

Remote access

Control the Pi from a laptop, tablet or phone with no monitor: at home with VNC or the web page, or from anywhere with Tailscale.

At home or on the hotspot: VNC

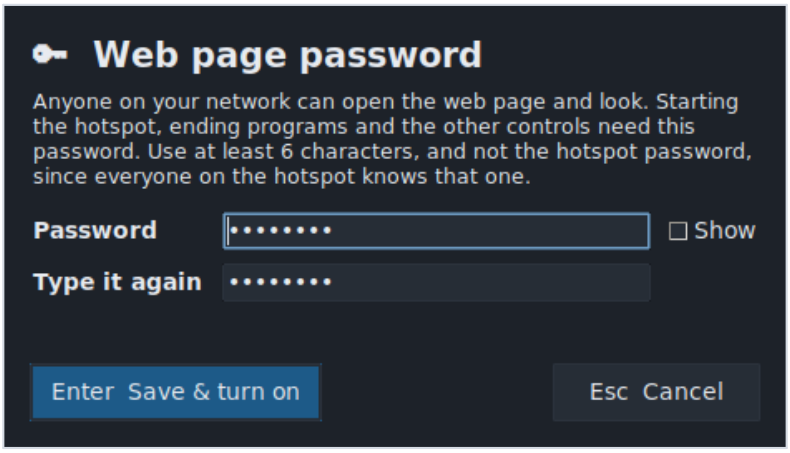
1. Install a VNC viewer app on the other device, such as **RealVNC Viewer** or **TigerVNC**.
2. Put both devices on the same network: your home Wi-Fi, or RPI500-OPEN when you're out.
3. In the viewer, connect to the Pi's address: the one shown in the Task Manager's top bar at home, or **192.168.1.250** on the hotspot. The port is **5900**, the standard one.
4. Sign in with your Pi username and password.

> **Tip.** Because the Pi creates RPI500-OPEN by itself whenever it can't find a network, you can always reach it: join RPI500-OPEN, then connect to **192.168.1.250**. If the screen looks too small or too big with no monitor attached, reinstall with a different `VNC_RESOLUTION`.

On your network: the web page

VNC shows the whole desktop, which is fiddly on a phone. The **Web page** switch in the Tools tab instead serves the Task Manager as a web page that fits any screen. Nothing needs installing on the phone or PC, just a web browser.

1. On the Tools tab, flip **Web page** on. The first time, it asks you to choose a **password** for the web page's controls (6 characters or more). Press `Enter` to save.
2. The line under the switch shows the address, for example **http://192.168.1.20:8090**. On the hotspot it's **http://192.168.1.250:8090**.
3. Open that address in a browser on any phone, tablet or PC on the same network. Add it to your home screen and it opens like an app.



Choosing the web page's password the first time you switch it on

On the web page	Who can use it
Gauges, processes, network lights, hotspot status	Anyone on your network, without logging in
Starting or stopping the hotspot, who's connected, the visitor log, ending programs, Bluetooth, CPU mode, overclock, stability test, Tailscale, reboot	Only after **Log in** with your web password

- It keeps running in the background, even when the Task Manager window is closed, and starts again after a reboot until you switch it off.
- The **Web page** light on the Overview tab shows when it's running.
- With Tailscale on as well, the web page also works from anywhere at **http://100.x.y.z:8090**.
- The AI chat, the Wi-Fi band switch and graphics boost stay in the Task Manager on the Pi itself.
- The visitor log on the web page is the same Desktop log the Task Manager keeps. It's written while the Task Manager is running.

Heads up. Anyone on your network, including people on the hotspot, can open the page and look. Pick a web password that's different from the hotspot password, since everyone on the hotspot knows that one.

From anywhere: Tailscale

Tailscale links your own devices into a private network that works over any internet connection, with no router setup. It's free for personal use. Once it's on, your phone on mobile data or your PC at a friend's house can reach the Pi exactly as if you were home.

1. On the Tools tab, flip **Tailscale** on. The first time, it installs itself, which needs internet for about a minute.

2. A window shows a QR code. Scan it with your phone's camera (or open the link), sign in with a Google, Microsoft, Apple or GitHub account, and press **Connect**. The window closes by itself once the Pi has joined.
3. Install the **Tailscale** app on your phone and PC, and sign in with the same account.
4. Use the Pi's Tailscale address, shown under the switch (it starts with **100.**). In your VNC viewer connect to **100.x.y.z:5900**, or use **ssh** to it. It works from anywhere with internet.



The Tailscale sign-in window. Scan the code with your phone to sign in.

- Switching Tailscale off keeps you signed in, so turning it back on is instant. It also stays on across reboots.
- Only devices signed in to **your** Tailscale account can reach the Pi this way, and VNC still asks for your Pi password.
- The Pi itself needs internet. While it's running RPI500-OPEN, that means it also needs an Ethernet cable to your router.

Tip. Setting up a Pi with no screen? In the Tailscale admin console, create an **auth key**, then install with `sudo TAILSCALE_AUTHKEY=tskey-… bash install.sh`. The Pi joins by itself, with no QR code needed.

CHAPTER 12

Running from a battery bank

Tips for taking the Pi 500+ out and about.

- **Use a USB-C Power Delivery bank** of at least 15 W (5 V 3 A), and ideally one that offers 5 V 5 A for full USB power, with a short, thick USB-C cable. Long or thin cables are the most common cause of low-voltage warnings.
- **Watch the Input voltage gauge** on the Overview tab. It shows the real volts arriving, and it sags as the bank runs down. The Power draw gauge shows how hard the Pi is working, and the Power supply line shows whether the bank gave 3 A or 5 A.
- **The Power light** on the Overview tab is a second warning. Red *Right now* means the bank or cable can't keep up. If it happens as the bank empties, it's time to charge.
- **Switch CPU mode to Battery saver** on the Tools tab to stretch runtime.
- **Turn the AI off** when you're not using it, and consider leaving the overclock at Stock.
- **Keep an eye on temperature**, especially in a bag or in the sun.

How long a bank lasts depends on its size and on how hard the Pi is working, so try yours at home first and note how long it runs.

CHAPTER 13

Keyboard shortcuts

Everything works from the Pi 500+'s own keyboard. No mouse needed.

Keys	Where	What it does
`Ctrl+Shift+Esc`	Anywhere	Open the Task Manager
`Ctrl+Alt+B`	Anywhere	Bluetooth pairing window, keyboard only, even with the app closed
`S` / `R`	Bluetooth window	Scan again / reconnect paired devices
`Enter` / `Esc`	Stability test	Start the test / stop it or close the window
`F1` – `F5`	Task Manager	Overview, Processes, Network, Tools, AI
`Alt+1` – `Alt+5`	Task Manager	Same as F1 to F5
`Ctrl+P`	Task Manager	Open Who's Connected
`Ctrl+H`	Task Manager	Start or stop the hotspot
`Ctrl+M`	Task Manager	Clean & optimize memory
`Ctrl+U`	Task Manager	Update everything
`Ctrl+B`	Task Manager	Bluetooth rescue
`Ctrl+F`	Task Manager	Search the process list
`Del` / `Shift+Del`	Processes tab	End task / force kill
`Ctrl+Q`	Task Manager	Quit
`Del`	Who's Connected	Disconnect the selected device
`Ctrl+S`	Who's Connected	Save a report to the Desktop
`Ctrl+A`	Who's Connected	Turn auto-save on or off
`Enter` / `Esc`	AI tab	Send your question / stop the answer
`Tab`, arrows, `Space`	Everywhere	Move around and press buttons or switches

App shortcuts pause while you're typing in a text box, so typing in the AI chat or search box never sets anything off by accident.

CHAPTER 14

Troubleshooting

Problem	Try this
A button says *the root helper isn't installed*, or asks for a password	Run `sudo bash install.sh` again from the rpi500-taskmgr folder.
The hotspot won't start	Set the Wi-Fi country: *sudo raspi-config → Localisation Options → WLAN Country*, then try again. The hotspot needs Raspberry Pi OS Bookworm or newer.
A phone or gadget can't see RPI500-OPEN	It may only support 2.4 GHz. Flip the **Wi-Fi band** switch on the Tools tab to 2.4 GHz.
Teams or other apps act offline while my phone is on the hotspot	Turn on **Phones keep their mobile data** in the Tools tab, then forget the hotspot on the phone and join it again. See chapter 9.
I forgot the hotspot password I set	It's shown in the Tools tab. Or, over SSH or a terminal, `sudo rpi500-helper ap-set --default` puts back RPI500-OPEN / fifty-seven57.
RPI500-OPEN didn't appear after power-on	Wait a full minute. It only starts if the Pi isn't on any other network, so unplug the Ethernet cable. Check the **Auto-hotspot at boot** light on the Overview tab.
My Bluetooth mouse or keyboard stopped working	Press `Ctrl+Alt+B`, then `R` to reconnect your paired devices, and wake the device with a key or click. If it still won't connect, put it in pairing mode, press `S`, choose it and press Enter.
The Power light is red, or Input voltage is amber or red	Use a shorter, thicker cable and a charger or bank port that gives at least 3 A, or charge the bank.
The Power supply line says 3 A and a USB drive misbehaves	Use the official 27 W supply, or a bank that offers 5 V 5 A, or power the drive separately.
The Pi won't start after overclocking	See the Heads up box in the Performance section of chapter 7.
The Pi froze or restarted during a stability test	That speed is too high for your board. Go back one step on the Tools tab and reboot.
The web page won't open on my phone	Check the phone is on the same network as the Pi, and that the address ends in **:8090**. The **Web page** light on the Overview tab should be green.
I forgot the web page password	Press **Password** next to the Web page switch and choose a new one.
Update everything says there's no internet	Connect the Pi to your normal Wi-Fi or an Ethernet cable (the hotspot has no internet), then press it again.
The update didn't finish	Press **Update everything** again. It picks up where it stopped. If it keeps failing, run `sudo apt full-upgrade` in a terminal to see the problem.

Problem	Try this
Tailscale won't switch on	The Pi needs internet for Tailscale. Check it's on a network with internet access (on the hotspot, plug in an Ethernet cable), then try again.
I can't reach the Pi over Tailscale	Make sure your phone or PC is signed in to the same Tailscale account, its Tailscale app is on, and the Pi's switch shows On with a 100. address.
The VNC screen is tiny, black or the wrong size with no monitor	Reinstall with `VNC_RESOLUTION=1280x720` (or your preferred size) right after sudo, then reboot.
The AI says it needs 64-bit	Reinstall Raspberry Pi OS (64-bit) with Raspberry Pi Imager.
The AI is slow	Use qwen2.5:1.5b or llama3.2:3b, set CPU mode to Full speed, and close big programs such as the web browser.
The AI download failed	Make sure the Pi is online, then press **Download** again.
The visitor log on the Desktop isn't updating	Check the Auto-save switch in the Who's Connected window, and that the Task Manager is running.
`Ctrl+Shift+Esc` or **`Ctrl+Alt+B`** do nothing	In the rpi500-taskmgr folder run `sudo bash setup-shortcuts.sh`, then log out and back in. It sets them up for labwc, Wayfire or the older X11 desktop, whichever your Pi uses.

CHAPTER 15

Where things are kept

For the curious, or for fixing things by hand.

File or folder	What it is
`/opt/rpi500-taskmgr/`	The app itself
`/usr/local/bin/rpi500-helper`	The small script that runs the admin-level buttons
`/usr/local/sbin/rpi500-fallback-ap`	Starts the hotspot at boot when there's no network
`/etc/NetworkManager/dispatcher.d/90-rpi500-led`	Flashes the power light while the hotspot is on
`/etc/rpi500-taskmgr.conf`	Hotspot settings: name and password (AP_SSID, AP_PASS), band (AP_BAND=bg for 2.4 GHz or a for 5 GHz), sharing the Pi's internet (AP_SHARE=1 or 0) and whether the power light and keyboard flash (LED_BLINK=1 or 0)
`/etc/NetworkManager/dnsmasq-shared.d/rpi500-taskmgr.conf`	Present while phones keep their mobile data: tells the hotspot not to offer a route to the internet
`~/.config/rpi500-taskmgr/settings.json`	Your auto-save choice and AI model
`~/Desktop/RPI500-OPEN visitor log.csv`	The running visitor log
`/boot/firmware/config.txt`	Where the overclock is written, between markers

Admin actions can also be run from a terminal, which is handy over SSH:

```
sudo rpi500-helper ap-up        # start the hotspot
sudo rpi500-helper ap-down      # stop it and rejoin your Wi-Fi
sudo rpi500-helper ap-band 5    # hotspot on 5 GHz (or 2.4)
sudo rpi500-helper ap-set 'My Car' 'new-password'   # rename (or: ap-set --default)
sudo rpi500-helper ap-data keep # phones keep mobile data (or: share)
sudo rpi500-helper vpn up       # Tailscale on (down | logout)
sudo rpi500-helper web on       # web page for phones and PCs on (or: off)
sudo rpi500-helper update start # update everything (progress:
/var/log/rpi500-update.log)
sudo rpi500-helper bt-rescue    # Bluetooth rescue
sudo rpi500-helper oc 3000      # CPU to 3.0 GHz (stock|2600|2800|3000)
sudo rpi500-helper gpu-boost on # graphics to 1000 MHz (on|off)
rpi500-taskmgr --pair           # open the Bluetooth pairing window
sudo rpi500-helper led test     # flash the power light for 6 seconds
sudo rpi500-helper ai off       # stop the AI engine
```